Cultural Infusion

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Collingwood VIC 3066

Attendance Sas

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Document Status: Draft

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Objective and Background

Opportunity Statement

Many academic institutions in Australia track attendance with dedicated tools, such as student NFC cards, that add a cost students don't always cover. When those tools aren't used, institutions often fall back on manual roll calls or sign-in sheets, which are repetitive, slow and prone to errors. A system that lets students check in with the phones they already own, using details already stored in the system, would remove that cost and make attendance faster and more reliable for students and staff.

Business Goal

Design, develop and test an automated attendance system that lets students record their own attendance at classes, lectures and meetings using their personal devices. The system will:

- check students in through the room's Bluetooth beacon, NFC tag or QR code, falling back automatically when one isn't available
- use details already stored in the system, so students never fill in the same form twice
- reduce check-in time and remove the need for students to buy additional hardware

- give lecturers live and historical attendance for each session and class, with the ability to correct records
- give administrators one place to manage periods, classes, schedules, users and room equipment

Elevator Pitch

Taking attendance shouldn't cost students money or cost lecturers class time. Our system turns the phone every student already carries into their attendance card. When they walk into class, the app checks them in through a Bluetooth beacon, an NFC tag, or a QR code in the room. There's no extra hardware for students to buy, no paper sign-in sheets, and no repeated forms. Lecturers see who's present in real time and can fix any record, and administrators manage everything from one place. Faster check-ins, accurate records, and lower costs for everyone.

Target Users

Academic institutions: universities, colleges, and training providers that need accurate attendance records without buying hardware for every student.
Students: they check in to their classes with their own phone and see their upcoming, ongoing, and past sessions along with their attendance history.
Lecturers: they see who is present in each session as it happens, track attendance percentages per session and per student, and correct records when needed.
Administrative staff: they set up and manage academic periods, classes, schedules, users and room equipment, and can view and correct attendance across the institution.

Product features

Functional requirements

Administration

FR-01: Admins can create, view, edit and delete academic periods, classes, class schedules, users, roles, permissions, locations and check-in devices (Bluetooth beacons and NFC tags).

FR-02: Admins can assign a lecturer to a class schedule and enrol students in it.

FR-03: Admins can schedule class sessions, and reschedule or cancel individual sessions.

Viewing classes and sessions

FR-04: Students and lecturers can see their own ongoing, upcoming and past sessions and classes. Admins can see all of them.

Attendance

FR-05: Lecturers and admins can see, for each session, the enrolled students, each student's attendance status and the session attendance rate.

FR-06: Lecturers and admins can see a student's attendance history and attendance rate for each class.

FR-07: Lecturers can edit attendance records for their own sessions, and admins can edit any record. Each change requires a reason and records who made it.

FR-08: Students can check in to a session during its check-in window using a Bluetooth beacon, NFC tag or QR code. The app tries each method in that order and falls back automatically when one isn't available.

Access

FR-09: Users sign in with their institutional email through a one-time link or code, with no password.

Non-functional requirements

NFR-01, Availability: The system is available at least 98% of the time, which allows about 14.6 hours of downtime per month.

NFR-02, Security: Every request is authenticated, and users can only access data their role and permissions allow. Data is encrypted in transit.

NFR-03, Extensibility: New roles and permissions can be added without code changes, beyond student, lecturer and admin.

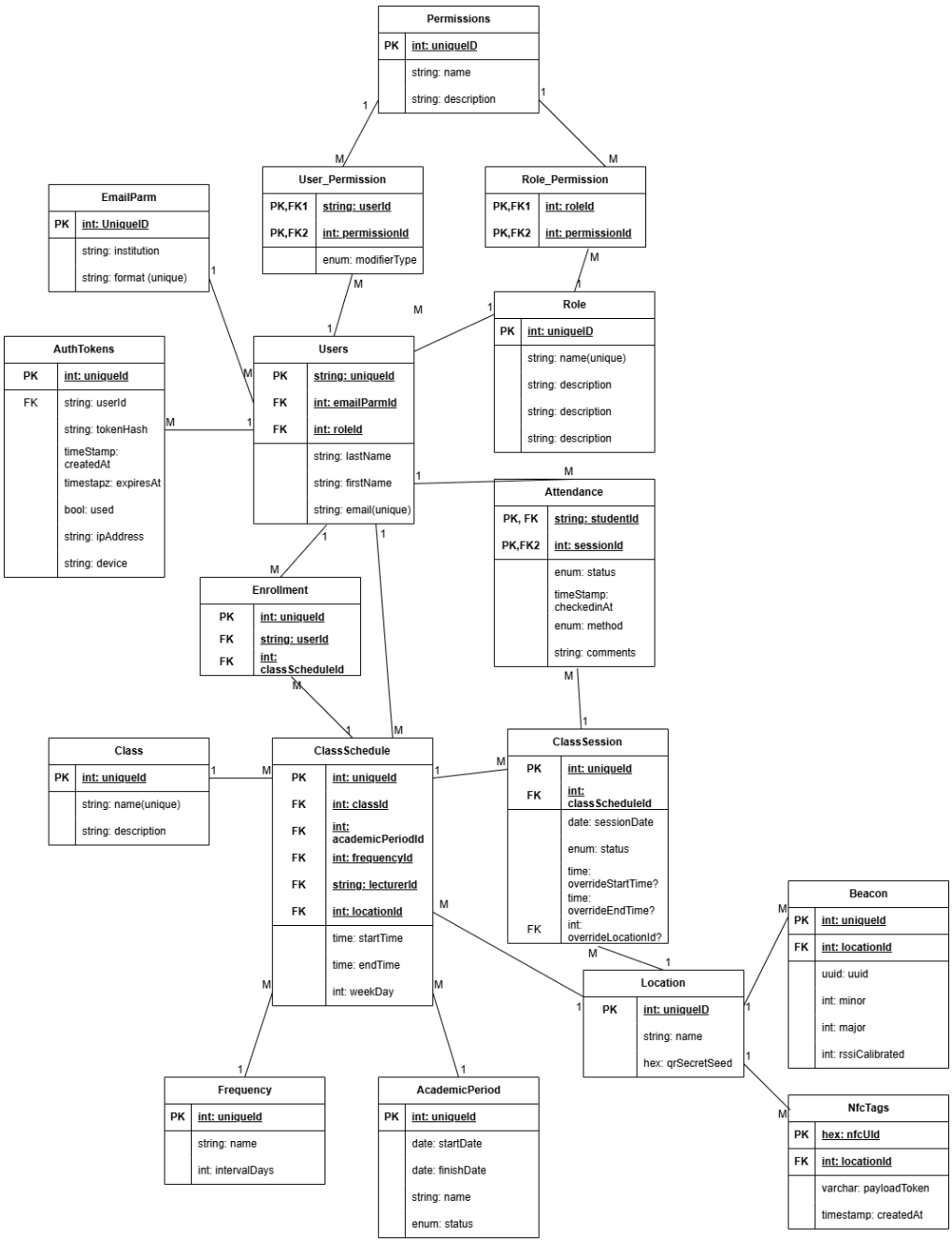
NFR-04, Modularity: The system can be extended with new modules, such as grading or document records, without changing the existing attendance features.

NFR-05, Performance: A check-in completes in under 5 seconds.

NFR-06, Privacy: Personal data is handled in line with the Australian Privacy Principles.

Design

ERD

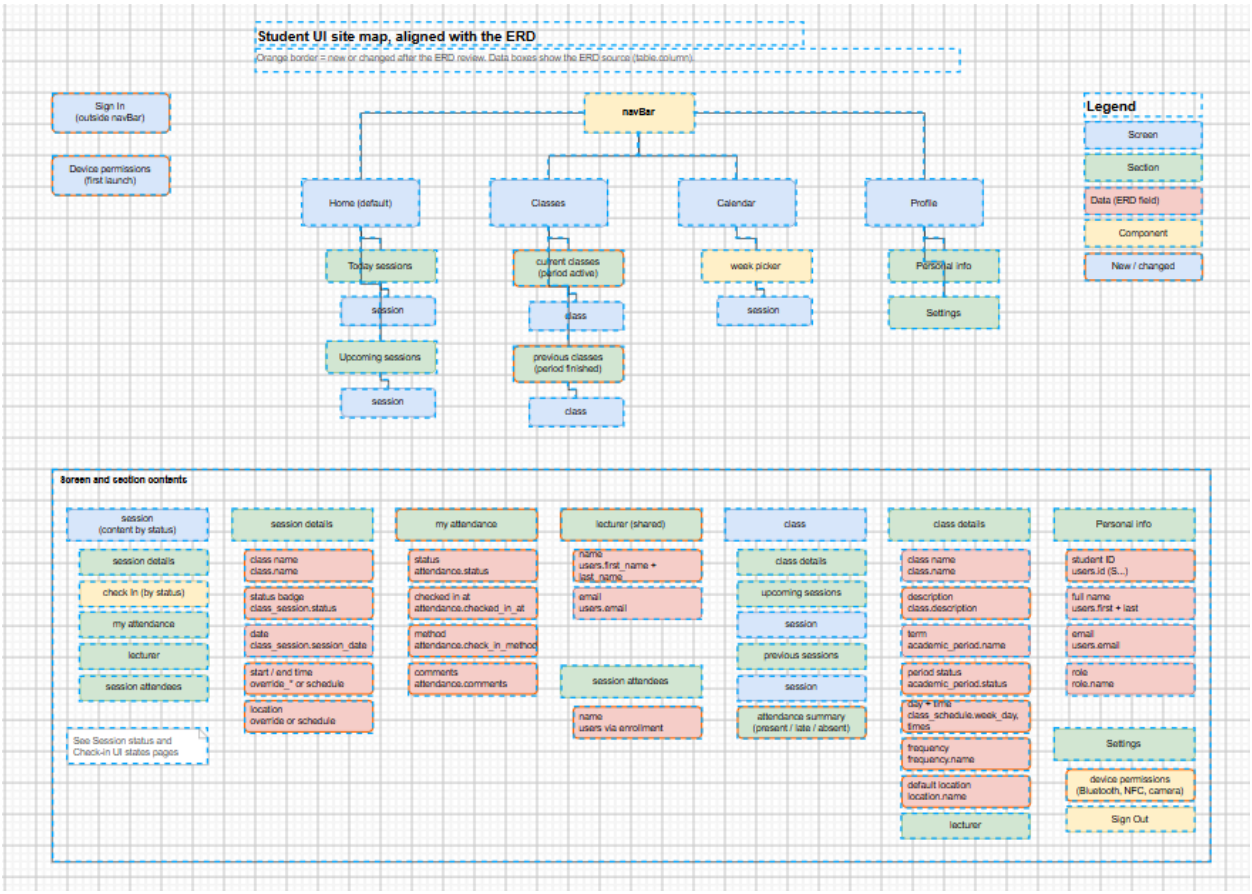

Permissions
PK int: uniqueID
string: name
string: description
User_Permission
PK,FK1 string: userId
PK,FK2 int: permissionId
enum: modifierType
Role_Permission
PK,FK1 int: roleId
PK,FK2 int: permissionId
EmailParm
PK int: UniqueID
string: institution
string: format (unique)
Role
PK int: uniqueID
string: name(unique)
string: description
string: description
string: description
AuthTokens
PK int: uniqueId
FK string: userId
string: tokenHash
timeStamp: createdAt
timestapz: expiresAt
bool: used
string: ipAddress
string: device
Users
PK string: uniqueId
FK int: emailParmId
FK int: roleId
string: lastName
string: firstName
string: email(unique)
Attendance
PK, FK string: studentId
PK,FK2 int: sessionId
enum: status
timeStamp: checkedinAt
enum: method
string: comments
Enrollment
PK int: uniqueId
FK string: userId
FK int: classScheduleId
Class
PK int: uniqueId
string: name(unique)
string: description
ClassSchedule
PK int: uniqueId
FK int: classId
FK int: academicPeriodId
FK int: frequencyId
FK string: lecturerId
FK int: locationId
time: startTime
time: endTime
int: weekDay
ClassSession
PK int: uniqueId
FK int: classScheduleId
date: sessionDate
enum: status
time: overrideStartTime?
time: overrideEndTime?
FK int: overrideLocationId?
Beacon
PK int: uniqueId
FK int: locationId
uuid: uuid
int: minor
int: major
int: rssiCalibrated
Location
PK int: uniqueID
string: name
hex: qrSecretSeed
NfcTags
PK hex: nfcUid
FK int: locationId
varchar: payloadToken
timestamp: createdAt
Frequency
PK int: uniqueId
string: name
int: intervalDays
AcademicPeriod
PK int: uniqueId
date: startDate
date: finishDate
string: name
enum: status

Data Dictionary

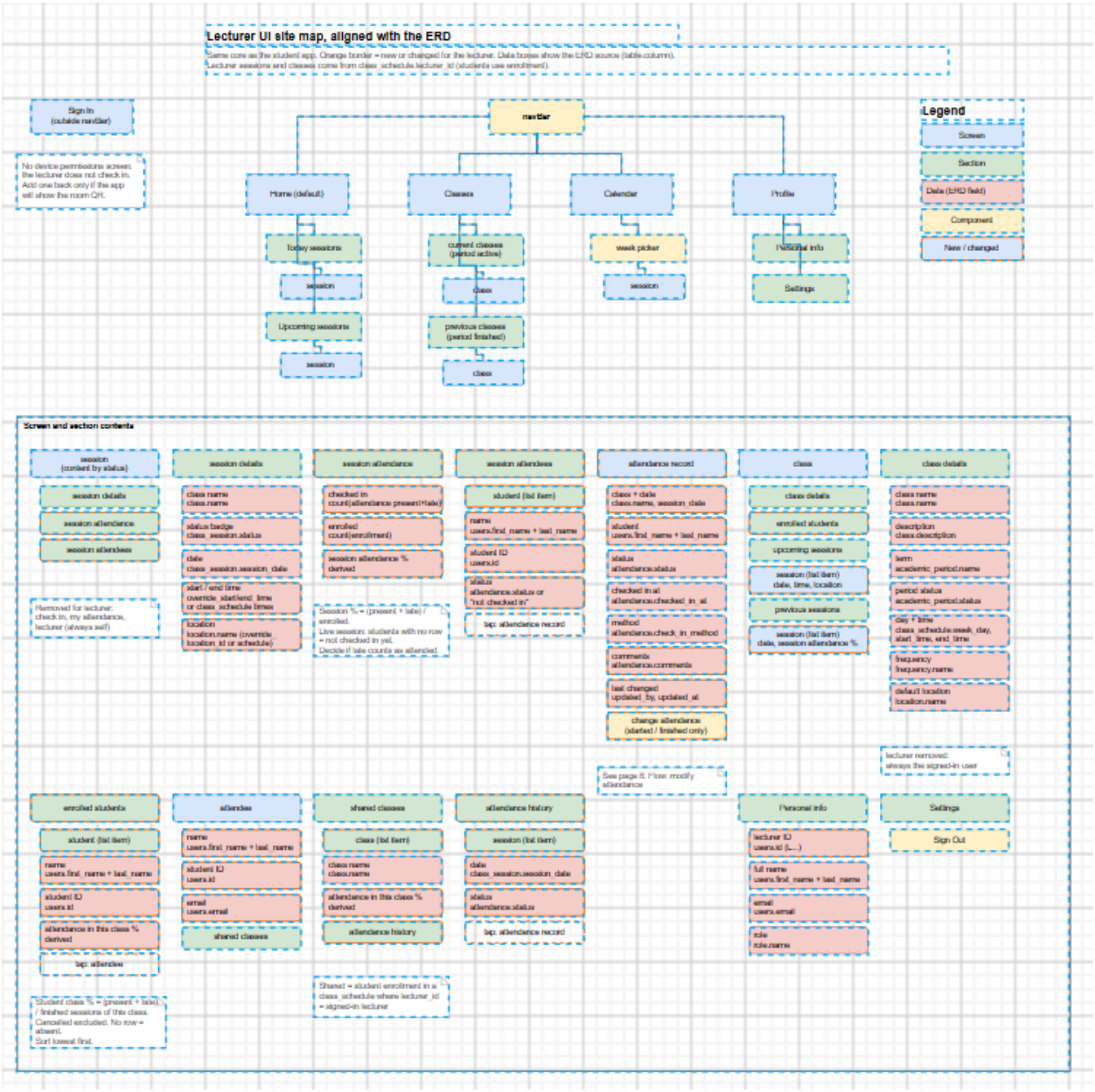
Data_Dictionary Attendace SAS

SiteMaps

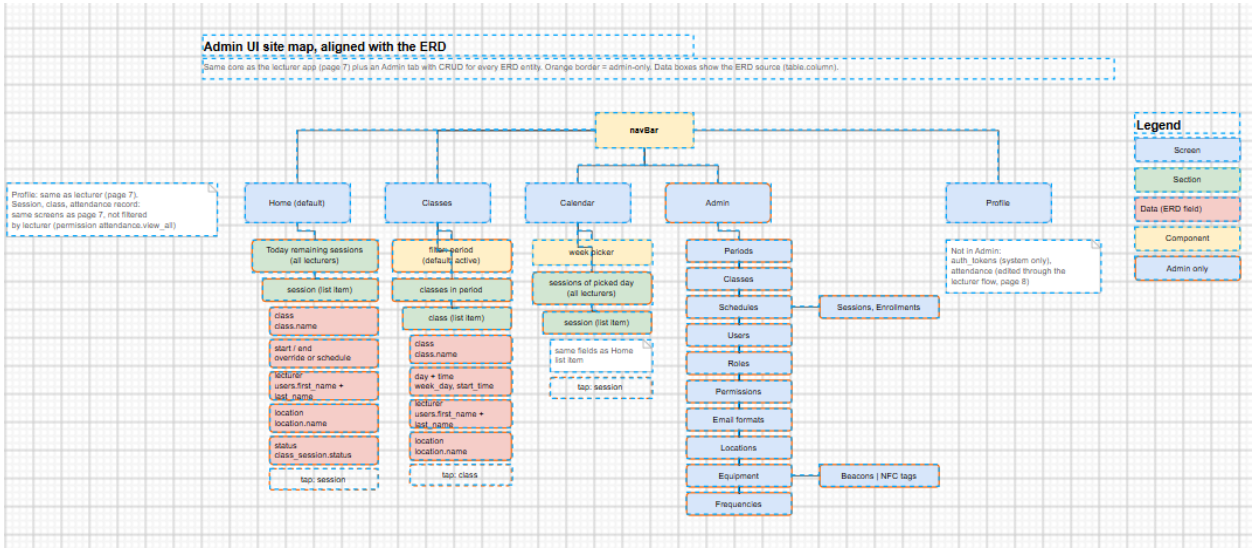
Student

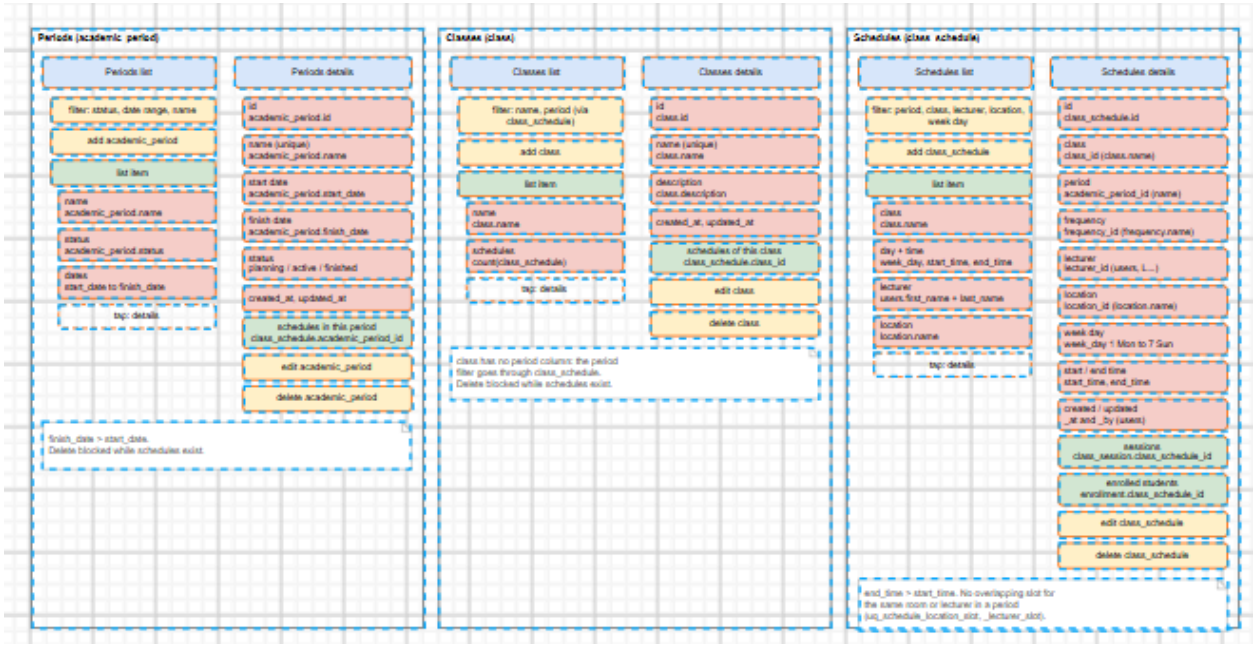


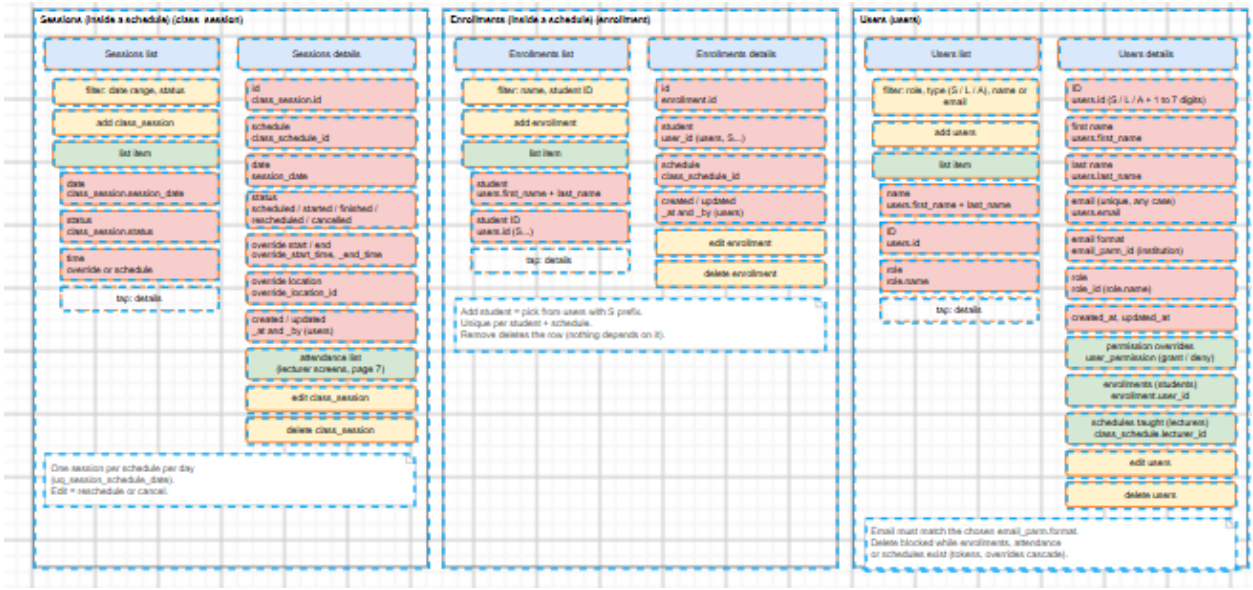
Lecturer

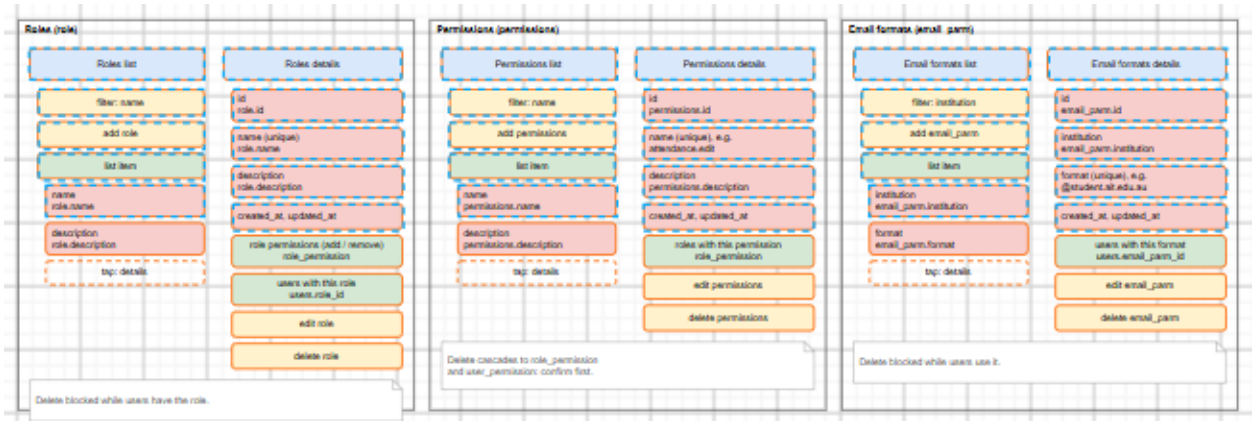

Lecturer UI site map, aligned with the ERD
Same core as the student app. Orange border = new or changed for the lecturer. Data boxes show the ERD source (table.column).
Lecturer sessions and classes come from class_schedule.lecturer_id (students use enrollment).
Sign In
(outside navBar)
No device permissions screen: the lecturer does not check in. Add one back only if the app will show the room QR.
navBar
Home (default)
Classes
Calendar
Profile
today sessions
session
Upcoming sessions
session
current classes
(period active)
class
previous classes
(period finished)
class
week picker
session
Personal info
Settings
Legend
Screen
Section
Data (ERD field)
Component
New / changed
Screen and section contents
session
(content by status)
session details
session attendance
session attendees
Removed for lecturer: check in, my attendance, lecturer (always self)
session details
class name
class.name
status badge
class_session.status
date
class_session.session_date
start / end time
override_start/end_time
or class_schedule.times
location
location.name (override
location_id or schedule)
session attendance
checked in
count(attendance.present+late)
enrolled
count(enrollment)
session attendance %
derived
Session % = (present + late) / enrolled.
Live session: students with no row = not checked in yet.
Decide if late counts as attended.
session attendees
student (list item)
name
users.first_name + last_name
student ID
users.id
status
attendance.status or
"not checked in"
tap: attendance record
attendance record
class + date
class.name, session_date
student
users.first_name + last_name
status
attendance.status
checked in at
attendance.checked_in_at
method
attendance.check_in_method
comments
attendance.comments
last changed
updated_by, updated_at
change attendance
(started / finished only)
See page 8: Flow: modify attendance
class
class details
enrolled students
upcoming sessions
session (list item)
date, time, location
previous sessions
session (list item)
date, session attendance %
class details
class name
class.name
description
class.description
term
academic_period.name
period status
academic_period.status
day + time
class_schedule.week_day,
start_time, end_time
frequency
frequency.name
default location
location.name
lecturer removed: always the signed-in user
enrolled students
student (list item)
name
users.first_name + last_name
student ID
users.id
attendance in this class %
derived
tap: attendee
Student class % = (present + late) / finished sessions of this class. Cancelled excluded. No row = absent.
Sort lowest first.
attendee
name
users.first_name + last_name
student ID
users.id
email
users.email
shared classes
shared classes
class (list item)
class name
class.name
attendance in this class %
derived
attendance history
Shared = student enrollment in a class_schedule where lecturer_id = signed-in lecturer
attendance history
session (list item)
date
class_session.session_date
status
attendance.status
tap: attendance record
Personal info
lecturer ID
users.id (L...)
full name
users.first_name + last_name
email
users.email
role
role.name
Settings
Sign Out

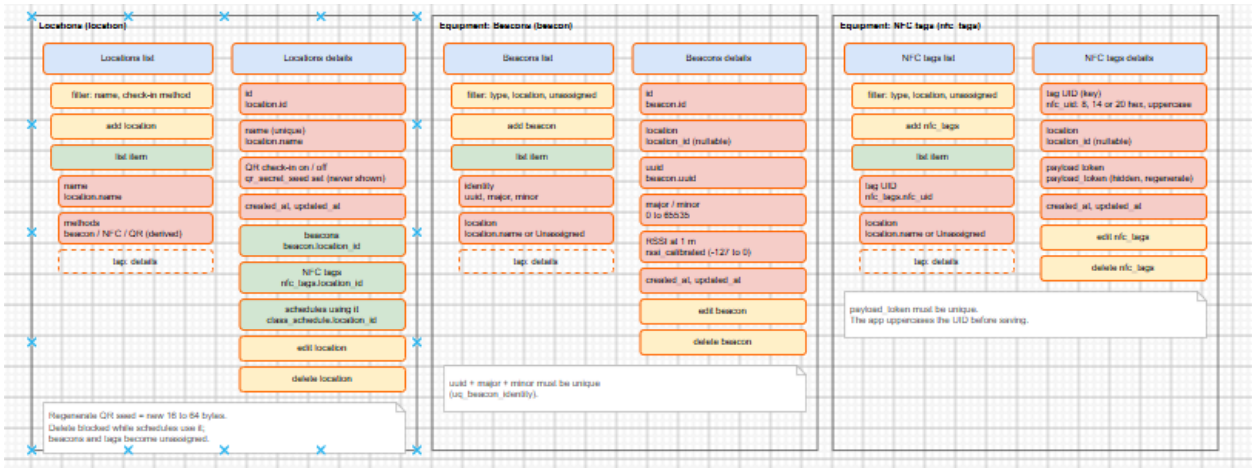
Admin

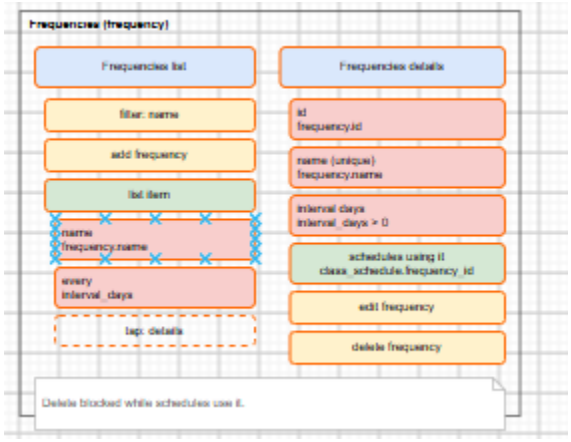

Admin UI site map, aligned with the ERD
Same core as the lecturer app (page 7) plus an Admin tab with CRUD for every ERD entity. Orange border = admin-only. Data boxes show the ERD source (table.column).
navBar
Legend
Screen
Section
Data (ERD field)
Component
Admin only
Profile: same as lecturer (page 7).
Session, class, attendance record:
same screens as page 7, not filtered
by lecturer (permission attendance.view_all)
Home (default)
Classes
Calendar
Admin
Profile
Today remaining sessions
(all lecturers)
session (list item)
class
class.name
start / end
override or schedule
lecturer
users.first_name + last_name
location
location.name
status
class_session.status
tap: session
filter: period
(default: active)
classes in period
class (list item)
class
class.name
day + time
week_day, start_time
lecturer
users.first_name + last_name
location
location.name
tap: class
week picker
sessions of picked day
(all lecturers)
session (list item)
same fields as Home
list item
tap: session
Periods
Classes
Schedules
Users
Roles
Permissions
Email formats
Locations
Equipment
Frequencies
Sessions, Enrollments
Beacons | NFC tags
Not in Admin:
auth_tokens (system only),
attendance (edited through the
lecturer flow, page 8)


Periods (academic_period)
Periods list
filter: status, date range, name
add academic_period
list item
name
academic_period.name
status
academic_period.status
dates
start_date to finish_date
tap: details
Periods details
id
academic_period.id
name (unique)
academic_period.name
start date
academic_period.start_date
finish date
academic_period.finish_date
status
planning / active / finished
created_at, updated_at
schedules in this period
class_schedule.academic_period_id
edit academic_period
delete academic_period
finish_date > start_date.
Delete blocked while schedules exist.
Classes (class)
Classes list
filter: name, period (via
class_schedule)
add class
list item
name
class.name
schedules
count(class_schedule)
tap: details
Classes details
id
class.id
name (unique)
class.name
description
class.description
created_at, updated_at
schedules of this class
class_schedule.class_id
edit class
delete class
class has no period column: the period
filter goes through class_schedule.
Delete blocked while schedules exist.
Schedules (class_schedule)
Schedules list
filter: period, class, lecturer, location,
week day
add class_schedule
list item
class
class.name
day + time
week_day, start_time, end_time
lecturer
users.first_name + last_name
location
location.name
tap: details
Schedules details
id
class_schedule.id
class
class_id (class.name)
period
academic_period_id (name)
frequency
frequency_id (frequency.name)
lecturer
lecturer_id (users, l...)
location
location_id (location.name)
week day
week_day 1 Mon to 7 Sun
start / end time
start_time, end_time
created / updated
_at and _by (users)
sessions
class_session.class_schedule_id
enrolled students
enrollment.class_schedule_id
edit class_schedule
delete class_schedule
end_time > start_time. No overlapping slot for
the same room or lecturer in a period
(uq_schedule_location_slot, _lecturer_slot).


Sessions (inside a schedule) (class_session)
Sessions list
Sessions details
filter: date range, status
add class_session
list item
date
class_session.session_date
status
class_session.status
time
override or schedule
tap: details
id
class_session.id
schedule
class_schedule_id
date
session_date
status
scheduled / started / finished / rescheduled / cancelled
override start / end
override_start_time, _end_time
override location
override_location_id
created / updated
_at and _by (users)
attendance list
(lecturer screens, page 7)
edit class_session
delete class_session
One session per schedule per day
(uq_session_schedule_date).
Edit = reschedule or cancel.
Enrollments (inside a schedule) (enrollment)
Enrollments list
Enrollments details
filter: name, student ID
add enrollment
list item
student
users.first_name + last_name
student ID
users.id (S...)
tap: details
id
enrollment.id
student
user_id (users, S...)
schedule
class_schedule_id
created / updated
_at and _by (users)
edit enrollment
delete enrollment
Add student = pick from users with S prefix.
Unique per student + schedule.
Remove deletes the row (nothing depends on it).
Users (users)
Users list
Users details
filter: role, type (S / L / A), name or email
add users
list item
name
users.first_name + last_name
ID
users.id
role
role.name
tap: details
ID
users.id (S / L / A + 1 to 7 digits)
first name
users.first_name
last name
users.last_name
email (unique, any case)
users.email
email format
email_parm_id (institution)
role
role_id (role.name)
created_at, updated_at
permission overrides
user_permission (grant / deny)
enrollments (students)
enrollment.user_id
schedules taught (lectures)
class_schedule.lecturer_id
edit users
delete users
Email must match the chosen email_parm.format.
Delete blocked while enrollments, attendance
or schedules exist (tokens, overrides cascade).

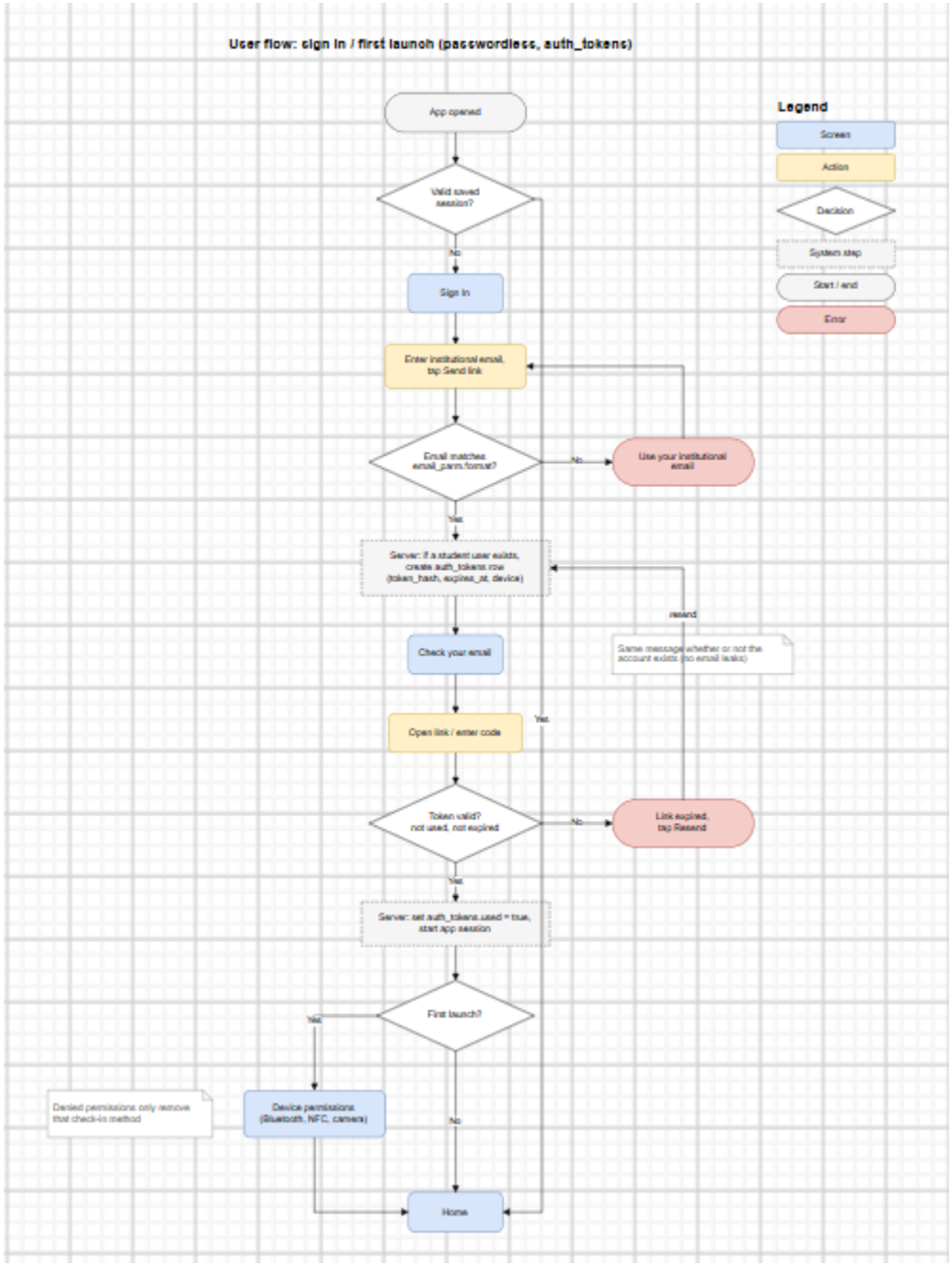

Roles (role)
Roles list
Roles details
filter: name
add role
list item
name
role.name
description
role.description
tap: details
id
role.id
name (unique)
role.name
description
role.description
created_at, updated_at
role permissions (add / remove)
role_permission
users with this role
users.role_id
edit role
delete role
Delete blocked while users have the role.
Permissions (permissions)
Permissions list
Permissions details
filter: name
add permissions
list item
name
permissions.name
description
permissions.description
tap: details
id
permissions.id
name (unique), e.g.
attendance.edit
description
permissions.description
created_at, updated_at
roles with this permission
role_permission
edit permissions
delete permissions
Delete cascades to role_permission
and user_permission: confirm first.
Email formats (email_parm)
Email formats list
Email formats details
filter: institution
add email_parm
list item
institution
email_parm.institution
format
email_parm.format
tap: details
id
email_parm.id
institution
email_parm.institution
format (unique), e.g.
@student.sit.edu.au
created_at, updated_at
users with this format
users.email_parm_id
edit email_parm
delete email_parm
Delete blocked while users use it.


Locations (location)
Locations list
Locations details
filter: name, check-in method
add location
list item
name
location.name
methods
beacon / NFC / QR (derived)
tap: details
id
location.id
name (unique)
location.name
QR check-in on / off
qr_secret_seed set (never shown)
created_at, updated_at
beacons
beacon.location_id
NFC tags
nfc_tags.location_id
schedules using it
class_schedule.location_id
edit location
delete location
Regenerate QR seed = new 16 to 64 bytes.
Delete blocked while schedules use it;
beacons and tags become unassigned.
Equipment: Beacons (beacon)
Beacons list
Beacons details
filter: type, location, unassigned
add beacon
list item
identity
uuid, major, minor
location
location.name or Unassigned
tap: details
id
beacon.id
location
location_id (nullable)
uuid
beacon.uuid
major / minor
0 to 65535
HSSI at 1 m
real_calibrated (-127 to 0)
created_at, updated_at
edit beacon
delete beacon
uuid + major + minor must be unique
(uq_beacon_identity).
Equipment: NFC tags (nfc_tags)
NFC tags list
NFC tags details
filter: type, location, unassigned
add nfc_tags
list item
tag UID
nfc_tags.nfc_uid
location
location.name or Unassigned
tap: details
tag UID (key)
nfc_uid: 8, 14 or 20 hex, uppercase
location
location_id (nullable)
payload token
payload_token (hidden, regenerate)
created_at, updated_at
edit nfc_tags
delete nfc_tags
payload_token must be unique.
The app uppercases the UID before saving.

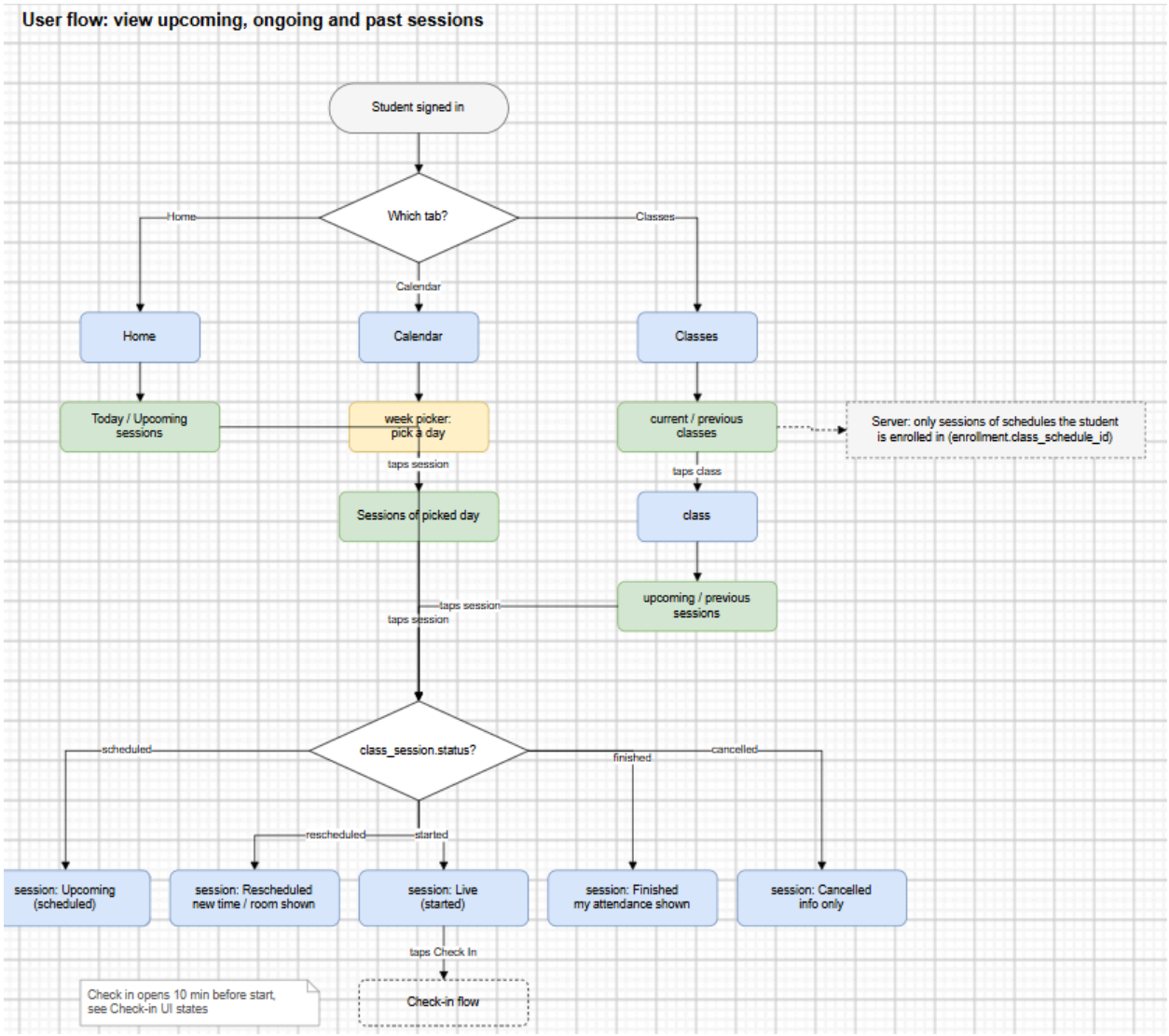

Frequencies (frequency)
Frequencies list
Frequencies details
filter: name
add frequency
list item
name
frequency.name
every
interval_days
tap: details
id
frequency.id
name (unique)
frequency.name
interval days
interval_days > 0
schedules using it
class_schedule.frequency_id
edit frequency
delete frequency
Delete blocked while schedules use it.

User Flow Diagram

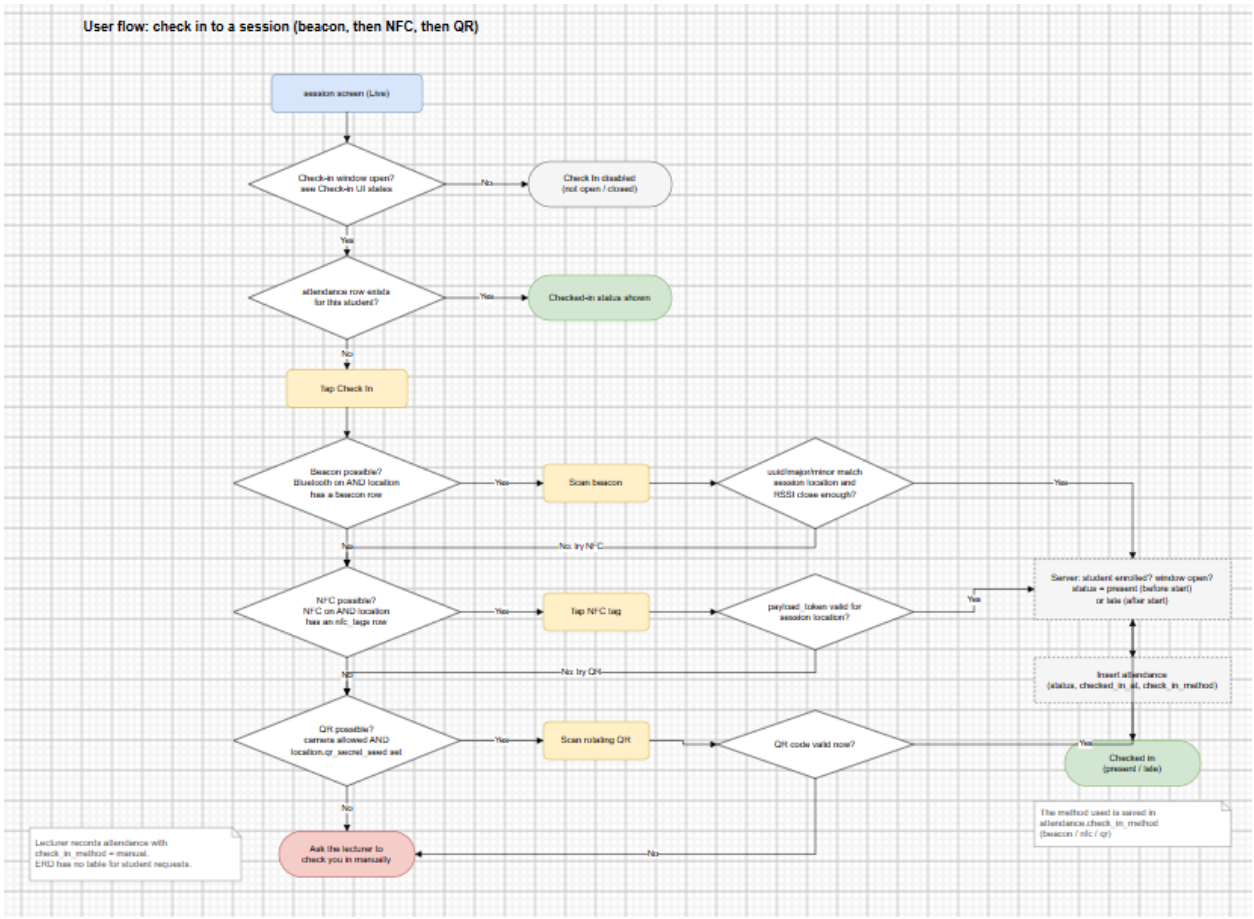
Sign In/ first Launch



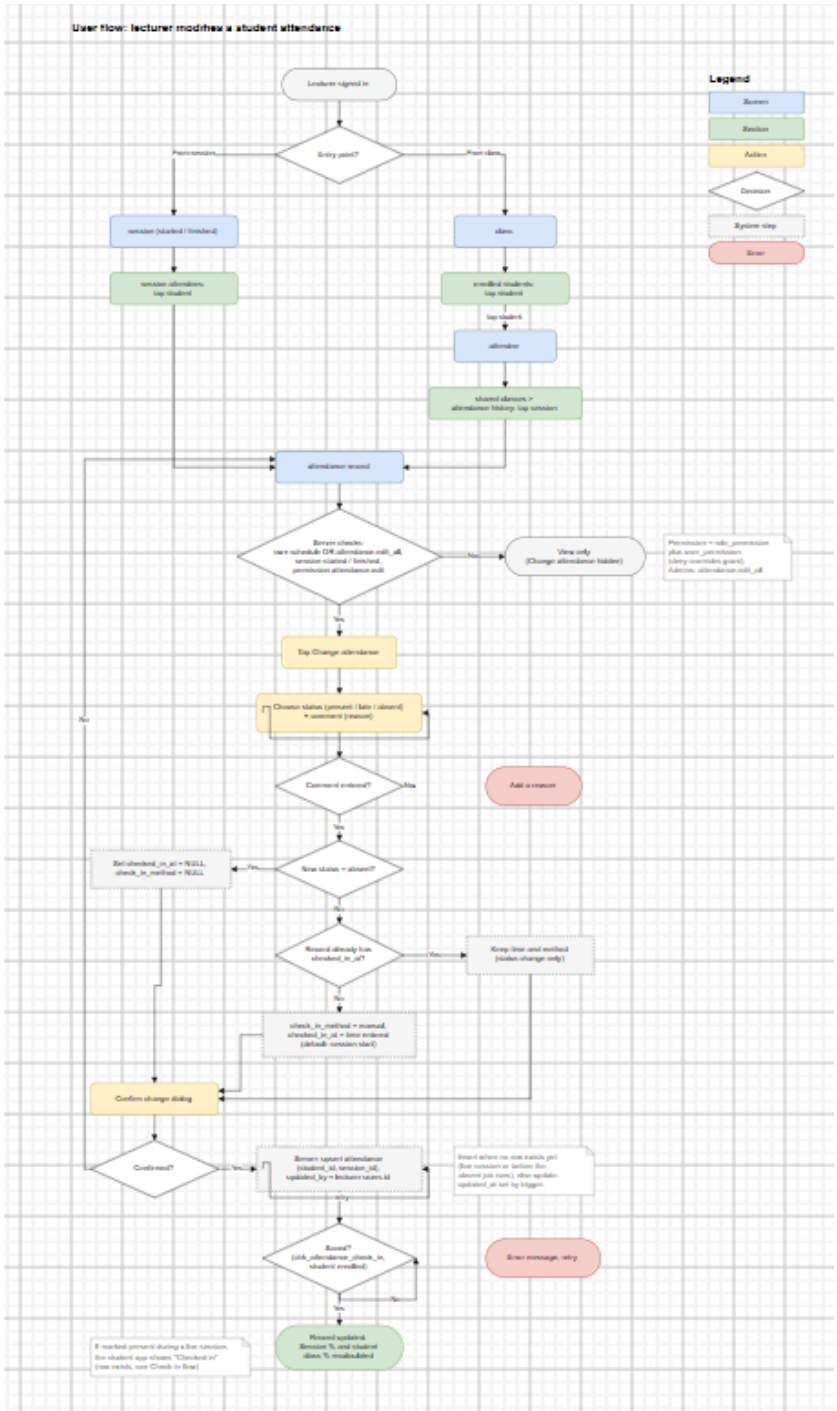
View Session



Check In


User flow: check in to a session (beacon, then NFC, then QR)
session screen (Live)
Check-in window open? see Check-in UI states
No
Check In disabled (not open / closed)
Yes
attendance row exists for this student?
Yes
Checked-in status shown
No
Tap Check In
Beacon possible? Bluetooth on AND location has a beacon row
Yes
Scan beacon
uuid/major/minor match session location and RSSI close enough?
Yes
No
No: try NFC
NFC possible? NFC on AND location has an nfc_tags row
Yes
Tap NFC tag
payload_token valid for session location?
Yes
Server: student enrolled? window open? status = present (before start) or late (after start)
No
No: try QR
Insert attendance (status, checked_in_at, check_in_method)
QR possible? camera allowed AND location.qr_secret_seed set
Yes
Scan rotating QR
QR code valid now?
Yes
Checked in (present / late)
No
The method used is saved in attendance.check_in_method (beacon / nfc / qr)
Lecturer records attendance with check_in_method = manual. ERD has no table for student requests.
Ask the lecturer to check you in manually
No

Modify Attendance



Wireframes:

Attendance SAS wireframes.pdf

Scope

In scope

- Admin management of all entities (FR-01 to FR-03)
- Session and class views for all three user types (FR-04)
- Attendance viewing and editing for lecturers and admins (FR-05 to FR-07)
- Student check-in with the three methods (FR-08)
- Passwordless sign-in (FR-09)

Out of scope

- Integration with third-party platforms such as learning management systems (for example, Canvas or Moodle)
- Grading and document records (the system is designed to support them later; see NFR-04)
- Buying and installing room equipment, which is the institution's responsibility